

1 Appendix: Extra experimental detail (slop warning)

1.1 Grouping, splits, and what is actually held out

We start with 11,797 reaction entries from the released Horizyn source data. Eighteen lack a named Rhea equation, leaving 11,779 entries for description preparation. “Clustering” here means connected components of exact-equivalence links, not a chemical-similarity clustering algorithm. Two reactions are linked if they share any of the following: (i) a canonical reaction SMILES, after removing atom-map numbers, canonicalizing each molecule with stereochemistry, sorting components within each side, and treating a reversal as equivalent; (ii) the same named equation after trimming and sorting its two sides; or (iii) the same Rhea master identifier. Transitive links are merged. This gives 11,461 groups, of which 11,335 are singletons; the largest contains 30 entries.

Each group’s ID is the SHA-256 hash of its sorted reaction IDs, joined by |. In the original paper dataset the split is determined by

$$u_g = \frac{\text{int}(\text{SHA256}(\text{horizyn-text-v1:} \parallel g)_{1:8}, 16)}{2^{32}}.$$

The thresholds are $u_g < 0.8$ for training, $0.8 \leq u_g < 0.9$ for validation, and $u_g \geq 0.9$ for test. All descriptions and equivalent reaction entries stay in the same split. The newer clean repository uses a different hash prefix, so rerunning its preparation command does *not* recreate this paper’s split; the frozen original manifests are needed.

Groups already used in the pilot validation/test sets were excluded from the fresh validation/test evaluation. The resulting cached dataset is:

Phase	Reaction IDs	Groups	Description rows
Train	9,397	9,152	46,845
Fresh validation	827	822	4,135
Fresh test	869	858	4,345
Pilot validation (excluded)	343	313	1,715
Pilot test (excluded)	334	307	1,670

Table 1: Exact counts in the frozen feature bundle used by the paper.

Training comprises 9,362 reactions with five styles each, plus 35 additional training reactions represented only by their canonical equation: 46,845 rows over 9,397 reaction IDs. Fresh validation has 827 reactions and fresh test has 869, each with all five styles. There are no shared exact-equivalence groups between these phases, and no exact train–fresh-test text matches after lowercasing and collapsing whitespace. Of the prepared non-training entries, nine lacked generated descriptions in the frozen snapshot and are absent from these evaluation counts.

This is an *adapter-held-out reaction-group split*. It is not a substrate scaffold, reaction-family, enzyme-family, protein-sequence, or temporal holdout. Related chemistry and proteins can occur on both sides, and the frozen Horizyn inference checkpoint already saw its released source data. For the later recipe, full-library contrastive and distillation losses also expose unlabelled held-out reaction vectors during training; their held-out descriptions are never training examples. These limits matter when interpreting “generalization.”

1.2 Retrieval protocol and the same-set teacher reference

Every test description searches the same 11,797 reaction entries or 216,132 protein entries by cosine similarity. Reaction positives are all entries in the query’s equivalence group. Protein positives are the union of annotations for that group in the released train- and test-pair files, restricted to the protein library. All

Query representation	Protein R@10 (%)	Protein R@100 (%)
True Horizyn reaction vector	62.49	78.71
4B language adapter (Table 1)	30.39	60.74
8B language adapter (Table 1)	32.60	61.66

Table 2: Same queries, protein candidates, and positive annotations for all rows.

vector for its five descriptions gives the identical rate over 4,345 queries. The 8B language adapter retains the following fraction of the teacher’s aggregate hit rate:

$$\frac{61.66\%}{78.71\%} = \mathbf{78.34\%}.$$

The absolute gap is 17.05 percentage points.

This ratio is not literally the fraction of teacher successes preserved: success sets differ. Conditional on the teacher succeeding, the language adapter succeeds on **72.40%** of queries, averaged over seeds. It also succeeds on **21.98%** of queries where the teacher fails. We therefore call this an *oracle-input teacher reference*, or an operational ceiling for reproducing the teacher, rather than a mathematical upper bound on every possible text query. The original paper’s “over 75%” result is not the denominator used here.

1.3 Performance by description style

The following breakdown uses the original saved heads behind Table 1, with three-seed means and 869 queries per individual style. “Plain + informal” contains 1,738 queries. All recall columns are percentages.

Encoder / style	MRR	Rxn R@10	Rxn R@100	Prot R@10	Prot R@100
4B / equation	0.2623	53.20	87.65	32.60	61.57
4B / precise	0.2656	52.21	87.23	32.18	62.75
4B / transformation	0.2378	48.68	83.47	29.77	61.22
4B / plain	0.2178	43.46	82.82	29.65	59.53
4B / informal	0.2097	45.88	81.40	27.73	58.61
4B / plain + informal	0.2137	44.67	82.11	28.69	59.07
4B / all	0.2386	48.68	84.51	30.39	60.74
8B / equation	0.2984	57.69	89.64	35.02	62.87
8B / precise	0.2754	57.12	89.72	34.56	63.25
8B / transformation	0.2684	53.01	86.80	31.03	62.18
8B / plain	0.2237	49.67	84.08	32.03	60.64
8B / informal	0.2370	49.98	83.24	30.34	59.38
8B / plain + informal	0.2303	49.83	83.66	31.19	60.01
8B / all	0.2606	53.49	86.70	32.60	61.66

Table 3: Description-style breakdown for the original Qwen3-Embedding scaling-study heads. Recall columns are percentages; MRR is reaction MRR.

For the headline 8B model, protein R@100 drops from 62.87% on equations to 59.38% on informal descriptions: a 3.49-point paired gap (95% group-bootstrap interval [0.97, 6.00] points). Plain/informal queries together achieve 60.01%, compared with 61.66% overall. The aggregate therefore gets a modest 1.65-point lift relative to these two styles; it does not conceal a collapse on informal text. Reaction retrieval shows a clearer specificity penalty: reaction R@100 is 89.64% for equations versus 83.24% for informal queries. These are synthetic descriptions of known reactions; style labels do not guarantee that names or

other identifying details are absent, and this does not replace evaluation on independently written human queries.

1.4 The catalogue baseline

The catalogue baseline uses the frozen Qwen3-Embedding-4B encoder to embed each candidate’s canonical named equation, with the same instruction and pooling as the description queries. The index contains 11,779 equations, including held-out reactions; all 869 test reaction IDs are covered. The remaining 18 reaction-library entries have no catalogue text. There is no learned adapter. Direct reaction retrieval ranks catalogue equations by $s_{ij} = x_i^\top c_j$. For protein retrieval we use the existing demo rule: take the five best equations, softmax their scores at temperature 0.03, and combine their Horizyn reaction vectors:

$$z_i^{\text{cat}} = \text{normalize} \left(\sum_{j \in \text{Top5}(s_i)} \frac{\exp(s_{ij}/0.03)}{\sum_{\ell \in \text{Top5}(s_i)} \exp(s_{i\ell}/0.03)} t_j \right).$$

The resulting vector searches the same protein library. No top- k or temperature search was performed for this appendix.

Catalogue query style	MRR	Rxn R@10	Rxn R@100	Prot R@10	Prot R@100
all	0.8684	95.37	99.33	55.67	75.51
equation	1.0000	100.00	100.00	61.10	78.94
precise	0.9558	99.65	99.88	58.23	77.33
transformation	0.8636	96.32	99.31	55.58	76.41
plain	0.7729	91.83	99.19	51.09	72.61
informal	0.7496	89.07	98.27	52.36	72.27
plain + informal	0.7612	90.45	98.73	51.73	72.44
non-equation	0.8355	94.22	99.17	54.32	74.65

Table 4: Qwen3-Embedding-4B catalogue baseline. Reaction scores use direct text matching; protein scores use the top-five vector mixture. Recall is in percent.

The catalogue achieves **75.51% protein R@100**, versus 60.74% for the 4B scaling-study adapter using the same encoder, and 78.71% for the true reaction input. Its plain/informal result is **72.44%**. The perfect equation reaction score reflects exceptionally strong matching cues in this setting; the non-equation results are the more informative baseline. This is a strong practical option for a known reaction inventory. It has access to each held-out candidate’s text, so it measures catalogue-assisted retrieval rather than learning to map previously unseen reaction descriptions without those candidate descriptions. Its quality on reactions missing from the catalogue is not established by this test.

1.5 Loss-component ablations

We ran every nonempty subset of the later adapter’s three losses, keeping the Qwen3-Embedding-4B features, training mean, centered linear architecture, data, optimizer, and validation-selection rule fixed. Retained coefficients stay at $(\lambda_{\text{cos}}, \lambda_{\text{contrast}}, \lambda_{\text{distill}}) = (1, 0.5, 1)$; removing a term sets its coefficient to zero. We did not retune the remaining coefficients or learning rate for individual subsets. Each configuration uses seeds 42, 43, and 44, with matched initialization and minibatch order for a given seed. The full model is the predeclared reference, not the winner of a test-set search. This is a retrospective ablation on an already-used test set, not a new blind benchmark.

Loss terms	Reaction MRR	Reaction R@100	Protein R@100
Cos + 0.5 Contr + Dist	0.2836 $\pm$ 0.0011	90.43 $\pm$ 0.01	65.75 $\pm$ 0.09
0.5 Contr + Dist	0.2846 $\pm$ 0.0003	90.95 $\pm$ 0.09	65.78 $\pm$ 0.15
Cos + Dist	0.2882 $\pm$ 0.0011	90.33 $\pm$ 0.01	65.28 $\pm$ 0.11
Cos + 0.5 Contr	0.2643 $\pm$ 0.0018	89.53 $\pm$ 0.04	65.94 $\pm$ 0.17
Cos only	0.1746 $\pm$ 0.0002	72.32 $\pm$ 0.06	51.67 $\pm$ 0.10
0.5 Contr only	0.2658 $\pm$ 0.0019	90.26 $\pm$ 0.07	66.34 $\pm$ 0.13
Dist only	0.2912 $\pm$ 0.0015	91.13 $\pm$ 0.08	65.33 $\pm$ 0.27

Table 5: Matched centered 4B loss ablations: mean and sample standard deviation over three seeds. Recall columns are percentages. Cos: cosine alignment; Contr: full-library contrastive loss; Dist: similarity distillation.

Removing cosine alignment raises protein R@100 by 0.03 percentage points (full minus ablated: -0.03; 95% paired interval $[-0.49, 0.43]$ points). Removing contrastive retrieval lowers protein R@100 by 0.48 percentage points (full minus ablated: +0.48; 95% paired interval $[0.01, 0.96]$ points). Removing similarity distillation raises protein R@100 by 0.18 percentage points (full minus ablated: -0.18; 95% paired interval $[-0.86, 0.47]$ points).

The effects on reaction ranking and protein retrieval need not agree: the checkpoint is selected for reaction MRR, while the downstream protein metric is a separate outcome. Small differences with intervals spanning zero do not establish that one objective is better. These are fixed-weight ablations, not separately optimized alternatives.

The predeclared full centered 4B recipe obtains 65.75% protein R@100, exceeding the uncentered 8B scaling-study result of 61.66%. This is a recipe comparison as well as an encoder comparison; it does not overturn or isolate the effect of encoder size.

For reaction MRR, the full objective scores 0.2836 versus 0.2643 without distillation (paired difference +0.0193; 95% interval $[0.0141, 0.0248]$). The evidence for distillation is stronger for reaction ranking than for downstream protein hits. This experiment does not establish that all three components are necessary. Cosine-only training reaches just 51.67% protein R@100. Distillation alone has the highest observed reaction MRR (0.2912), while contrastive loss alone has the highest observed protein R@100 (66.34%). These point estimates describe the sweep; they were not used for further tuning or test-set model selection.

The table reports mean $\pm$ sample standard deviation over the three seeds. That standard deviation describes training variation; it is not a test-set confidence interval. Paired intervals resample 858 whole test reaction groups 3,000 times (bootstrap seed 20260905), retain all correlated styles within each group, and average the three fixed seeds before resampling. They describe test-group uncertainty conditional on these runs, not all sources of model selection or annotation uncertainty.

1.6 Reproduction settings and compute

All linear heads discussed here use AdamW with learning rate 0.002, weight decay 0.01, batch size 256 (including the last partial batch), default $\beta = (0.9, 0.999)$ and $\epsilon = 10^{-8}$, and a cosine learning-rate schedule to zero. Text features and teacher vectors are L2-normalized; the later recipe also subtracts the mean of training description vectors and normalizes again. The center is a fixed buffer, not a trainable parameter. Temperatures are 0.07 for contrastive retrieval and

0.05 for teacher and student distillation distributions. There is no T^2 multiplier on the KL term.

For Table 1’s scaling-study heads, training runs for 48 epochs (8,784 optimizer steps), with validation every four epochs and no early stopping. The highest validation reaction MRR selects the checkpoint. Linear versus two-layer MLP architecture was also selected by mean validation MRR over seeds; all seven listed models selected linear. The comparison MLP used a 1,024-unit GELU hidden layer, dropout 0.1, and learning rate 0.001. The 4B linear checkpoints were selected at epochs 44/40/40 and the 8B ones at 48/44/40 for seeds 42/43/44, respectively. The table reports the three-seed mean, not the best seed’s test score.

The centered loss ablations use up to 64 epochs (11,712 steps), validation at epoch 1 and every fourth epoch, and early stopping after six consecutive validation checks without a strict MRR improvement. Equal MRR keeps the earlier checkpoint. All checkpoints use validation reaction MRR, never test or protein retrieval, for selection. All 21 loss-ablation runs completed 64 epochs. The full-objective checkpoints were selected at epochs 56/60/60 for seeds 42/43/44.

The 4B encoder supplies 2,560-dimensional features, so a biased linear head has $2560 \times 512 + 512 = \mathbf{1,311,232}$ trainable parameters: 1,310,720 weights and 512 biases. The 8B encoder supplies 4,096-dimensional features, giving $4096 \times 512 + 512 = \mathbf{2,097,664}$ parameters. The 1.31M claim therefore applies to the 4B adapter, not the headline 8B result.

Measured 4B configuration	Seconds per fit (three seeds)	Mean minutes
Full centered objective, up to 64 epochs	41.1, 42.4, 40.7	0.69
Scaling recipe, 48 epochs	24.0, 24.3, 24.1	0.40

Table 6: New synchronized per-fit measurements after embedding caches were loaded.

These new timings were measured on one NVIDIA A100-SXM4-80GB using PyTorch 2.6.0+cu124 and float32 head training with **high** matrix-multiplication precision. GPU synchronization brackets the fit timer. Times include optimization, scheduled validation, and copying the best state in memory; they exclude embedding generation, container startup, transfers, final test evaluation, and output serialization. The 24 fits took 12.64 minutes in total; the GPU function including reference/test evaluation and bookkeeping took 13.09 minutes. These are measured runtimes, not a dollar-cost estimate.

Final test ranking for the new ablations uses full float32 matrix multiplication with TF32 disabled, while preserving the fitted weights, selected epochs, and projected queries. This extra inference-only pass took 10.2 seconds. Compared with the initial fast evaluation, it changed 5 of 104,280 protein top-100 outcomes (24 heads times 4,345 queries); these small cutoff effects do not change the interpretation. Original Table 1 numbers are retained from their saved per-query ranks. An independent CPU top- k audit checks queries near both cutoffs as well as queries spread through the test set, including exact cutoff ties under the stated optimistic convention. All 3,048 query/cutoff checks passed.

Historical records also support the scale of the claim: the later selected 4B fit logged 46.11 seconds on a Modal L4, while the original 4B scaling sweep logged 93.35 seconds for six heads (linear and MLP, three seeds) after cached features were supplied. The 8B sweep logged 125.36 seconds for six heads on an A100-80GB. Those older timers have broader and different scopes and should not be presented as per-head timings. “Minimal compute” here means the incremental

adapter fit after features are cached; it excludes training the frozen encoders, description generation, and the initial text/protein/reaction embedding passes.

For Qwen3-Embedding-4B, the frozen revision is 5cf2132abc99cad020ac570b19d031efec650f2b, with last-token pooling, maximum length 1,024, and the prompt **Instruct: Represent this biochemical reaction for retrieving matching reactions and enzymes.** followed by a newline and **Query: <description>**. Features were normalized, cached in float16, then converted to float32 and normalized for fitting. The 8B Table 1 result uses saved 4,096-dimensional OpenRouter/Nebius embeddings of that same prompt; an API model name does not pin provider weights, so the audited cached vectors and their hashes are the exact reproduction artifact.

2 An 8B headline run, the paper recipe, and matched baselines

We kept the Qwen3-Embedding-8B encoder and the Horizyn libraries frozen and trained only on reaction targets. The evaluation uses the same 4,345 descriptions of 869 reactions (858 equivalence groups) as the scaling table. There are 216,132 candidate proteins and 11,797 reaction vectors. Here, $R@k$ means that at least one protein with a known matching reaction appears within the top k , with optimistic handling of exact score ties. Each reaction contributes all five description styles.

What was selected. The selected representation is a normalized equal-vector ensemble of 9 separately trained heads. We also report the best single head, the paper’s written recipe, and a catalogue baseline using the same 8B text encoder. Hyperparameters and checkpoints were selected using the separate validation split; the final test inputs were not mounted in the training or selection jobs. “Best” means best validation score among this explicitly bounded search, not a claim of global optimality.

@Xrrrr@					
		Reaction Plain + informal			
Method	MRR	R@10 (%)	R@100 (%)	R@100 (%)	R@100 (%)
8B scaling recipe (previous)	0.2606	32.60	61.66	60.01	
Paper recipe, 8B (3-seed mean)	0.2966	37.92	66.44	64.69	
Best single 8B head	0.2792	37.58	67.69	65.30	
Selected 8B adapter	0.2876	38.87	67.89	65.71	
8B catalogue	0.8800	56.71	75.63	72.44	
4B catalogue (previous)	0.8684	55.67	75.51	72.44	
True Horizyn reaction query	—	62.49	78.71	78.71	

Table 7: Identical held-out queries and protein candidate pool. Three-seed means average metrics from three separate heads; they are not ensembles. The selected row is a single deployable query representation, whose construction is stated above. Reaction MRR for catalogues measures direct text-to-canonical-equation retrieval; the teacher reaction metric is omitted because its query is the true reaction vector itself.

The selected adapter’s protein $R@100$ is 67.89% [65.02, 70.78], compared with 75.63% [73.06, 78.13] for the 8B catalogue and 78.71% [76.07, 81.38] for the true Horizyn reaction query. The paired adapter-minus-catalogue difference is -7.73 percentage points (95% interval -10.80 to -4.77). Intervals resample the

858 whole reaction-equivalence groups 3,000 times, preserving their descriptions; fixed-seed means are averaged within each sampled group.

The paper’s recipe as written. This is the centered linear adapter from Section 2.2:

$$u_i = \text{normalize}(x_i - \mu), \quad z_i = \text{normalize}(Wu_i + b), \quad \mathcal{L} = \mathcal{L}_{\text{cos}} + 0.5\mathcal{L}_{\text{contrast}} + \mathcal{L}_{\text{distill}}.$$

The full reaction library supplies the contrastive candidates, equivalent reactions are multi-positive targets, and temperatures are 0.07 for contrastive retrieval and 0.05 for distillation. The batch size is 256 and the optimizer is AdamW. For 8B the necessary width change is $4096 \rightarrow 512$, replacing the draft’s $2560 \rightarrow 512$ 4B example. We did not silently substitute the earlier scaling recipe.

The draft omits several training knobs. We filled these from the corresponding repository defaults: learning rate 0.002, weight decay 0.01, cosine learning-rate decay over at most 64 epochs, validation reaction MRR checked at epoch 1 and every 4 epochs, and early stopping after 6 unsuccessful checks. Seeds are 42, 43 and 44. The selected epochs were 44, 48, 56. This row averages all three seeds, giving 66.44% protein R@100; it is not the most favorable test seed.

2.1 Results by description style

Query style	Paper recipe	Protein R@100 (%)		
		Best single	Selected	8B catalogue
Equation	68.55	70.77	70.89	78.83
Precise	66.74	68.81	68.35	77.68
Transformation	67.51	68.24	68.81	76.75
Plain	65.59	65.13	66.05	71.92
Informal	63.79	65.48	65.36	72.96
Plain + Informal	64.69	65.30	65.71	72.44
All	66.44	67.69	67.89	75.63

Table 8: Each individual style has 869 queries; plain + informal has 1,738. The true Horizyn query has 78.71% R@100 for every style because its reaction vector does not depend on the description. Full reaction and protein metrics, including R@10, are supplied in the accompanying CSV.

The plain/informal average is 65.71% for the selected adapter and 72.44% for the catalogue. These are the most relevant rows for the informal-search claim. Equation and informal scores should be reported separately rather than treating the aggregate as a measurement of informal language alone. The five descriptions are synthetic variants of the same reaction facts, not independently collected user queries.

2.2 Which Horizyn misses does language recover?

For this exact test set, querying with the true Horizyn reaction vector retrieves a known positive protein within 100 hits for 684/869 reactions, or 78.71%. That leaves 185 missed reactions, replicated across their five descriptions: 925 failed description-level queries. Define

$$\text{rescue@100} = \frac{\#\{q : r_T(q) > 100 \wedge r_L(q) \leq 100\}}{\#\{q : r_T(q) > 100\}}.$$

The selected adapter recovers 368/925 of those description-level misses: **39.78%** [34.23, 45.33]. On plain/informal descriptions it recovers 134/370 (36.22%). These are conditional rescue rates, not percentage-point gains in overall recall.

Query style	Horizyn misses (queries)	Adapter rescues (queries)	Rescue rate (%)	
			Adapter	Catalogue
Equation	185	81	43.78	3.24
Precise	185	71	38.38	5.95
Transformation	185	82	44.32	11.89
Plain	185	66	35.68	11.35
Informal	185	68	36.76	14.59
Plain + Informal	370	134	36.22	12.97
All	925	368	39.78	9.41

Table 9: Both systems are compared with the same true-vector Horizyn misses. Catalogue rescue uses text retrieval followed by a mixture of Horizyn reaction vectors.

The full overlap for the selected adapter is: both succeed 2582 queries; adapter alone succeeds 368; Horizyn alone succeeds 838; both fail 557. Thus the adapter preserves 75.50% of the teacher’s successful queries. Its overall hit rate divided by the teacher’s hit rate is 86.26%; this different quantity includes both losses and rescues.

At the reaction level, at least one of the five descriptions rescues 110/185 teacher-missed reactions; using just plain or informal gives 88/185. These provide five and two retrieval opportunities respectively and must not be presented as single-query R@100. The teacher is an exact-test reference for the language bottleneck, not a mathematical upper bound: changed query geometry can recover annotated positives the true reaction vector misses. This is evidence of complementary retrieval, not proof of new biological activity or superiority on unseen reaction families.

2.3 Search, reproduction knobs and compute

The search screened 22 recipes with seed 42: centered and uncentered linear heads, loss-component weights and temperatures, two weight-decay settings, 1,024-hidden-unit GELU MLP and residual heads, and two ridge-regression controls. Gradient fits used up to 96 epochs, batch size 256, AdamW, cosine learning-rate decay, validation at epoch 1 and every 8 epochs, and 6 unsuccessful checks of patience. Linear heads used learning rate 0.002 and weight decay 0.01 unless the recipe explicitly used 0.1; MLP and residual heads used learning rate 0.001 and dropout 0.1. Exact settings for every run are saved in the experiment plan.

The three highest-scoring non-ridge recipes were retrained with seeds 42/43/44 for up to 128 epochs. Final candidates were the 31 individual search heads, three within-recipe equal-vector ensembles, and one ensemble of all nine refined heads. Selection maximized validation protein R@100, with ties broken by protein R@10, protein MRR, then reaction MRR. Protein annotations were used for this validation criterion only; no protein pair supplied a training gradient. The paper-recipe control retained its separate reaction-MRR selection criterion and was not added to the optimized search.

@Xrrr@				
Screen recipe	Cos., contr., distill.	Best epoch	Val. protein R@100 (%)	
	linear_cos_contrast	1, 1, 0	56	69.24
	linear_cd025	0, 1, 0.25	56	69.19
	linear_contrast_uncentered	0, 1, 0	88	69.17
	linear_contrast	0, 1, 0	56	69.12
	residual_contrast	0, 1, 0	56	69.07
	linear_cd1	0, 1, 1	40	68.97
	linear_contrast_wd01	0, 1, 0	72	68.90
	mlp_contrast	0, 1, 0	64	68.80
	linear_contrast_t010	0, 1, 0	64	68.71
	residual_full	1, 0.5, 1	56	68.68

Table 10: Ten highest-scoring screen recipes. These are validation results, not additional test comparisons.

The best single head was **linear_contrast_uncentered_refine_44**: a linear head with bias, with loss weights 0.0, 1.0, 0.0, contrastive temperature 0.07, distillation temperature 0.05, learning rate 0.002, weight decay 0.01, input centering disabled, selected epoch 64 (11,712 checkpoint updates); training stopped at epoch 112 (20,496 total updates). Its fit took 45.0 seconds on an NVIDIA A100-SXM4-80GB after embeddings were cached, including the scheduled validation and final validation pass. The three paper-recipe fits took 55.5, 42.5, 38.4 seconds (mean 45.5 seconds).

A $4096 \rightarrow 512$ linear head has 2,097,152 weights plus 512 biases: **2,097,664 trainable parameters**. The 1,311,232-parameter count belongs to the 4B encoder’s $2560 \rightarrow 512$ head. The selected representation contains 18,878,976 trainable parameters across 9 head(s); its members’ individual fit times sum to 457.0 seconds. Searching 22 configurations, refining 9 heads and running the 3 paper controls took 25.56 summed A100 fit-minutes, including validation. These timings exclude cached text/reaction/protein embedding creation, data upload, container startup, ensemble selection and final test evaluation. The encoder stays frozen; “8B” describes the encoder, not the number of trained parameters.

Catalogue details. The baseline embeds 11,779 canonical reaction equations with the same Qwen3-Embedding-8B model, instruction prefix and Nebius

provider as the queries. It retrieves the top 5 equations by cosine similarity, weights their Horizyn reaction vectors with $\text{softmax}(s/0.03)$, and normalizes the weighted sum before protein retrieval. These settings were inherited from the earlier catalogue baseline and held fixed for this experiment, with no further test-based tuning. Every test reaction is represented in the catalogue. No adapter is trained, but this is a closed-catalogue method with access to canonical test-reaction equations at inference.

All canonical vectors were refreshed together. Twelve repeated training-text anchors had cosine similarity 0.999869–0.999934 to the original query cache: close, but not bit-identical, so we did not mix freshly computed and old candidate vectors. The frozen query cache remained unchanged for every method. Cached float32 text features are normalized, rounded to float16, then converted back to float32 and normalized before adaptation. Training uses PyTorch’s high matmul precision; selection and test retrieval use highest float32 precision with TF32 disabled and CPU-normalized candidate libraries.

2.4 Limits and concrete recoveries

Limits of this comparison. The reaction split holds out exact-equivalence connected components, not chemical families, scaffolds or protein families. All five descriptions of a reaction share a split; there are 46,845 training and 4,135 validation rows. The full reaction candidate library is available during reaction-target training, while held-out descriptions and their target associations are not. The frozen Horizyn teacher and known-positive annotations originate in its released source data, so this does not constitute a new independent teacher test. Annotation incompleteness can count plausible proteins as misses. Finally, the test set has already appeared in the scaling analysis; these choices were frozen before this new test evaluation, but the study remains retrospective. Independent chemistry-family or user-query tests would be needed for stronger generalization claims.

Concrete plain/informal recoveries. The examples below are the first five distinct reactions when rescued plain/informal queries are sorted by smallest adapter rank, then largest teacher rank. Ranks refer to the best annotated positive for each method, which need not be the same protein. The listed protein is the adapter’s highest-scoring annotated positive. These are benchmark recoveries, not new biological validations.

Rh_30019: informal. “Cells decorate their outer coat by hooking a heptose sugar onto the lipid A-linked core, with ADP left behind.”

Best-positive ranks: adapter 1; true Horizyn 3240; 8B catalogue 7. Adapter protein: P27242.

Rh_43260: informal. “Chop the phosphate tag off a phosphatidic acid and you get a diacylglycerol plus free phosphate.”

Best-positive ranks: adapter 1; true Horizyn 272; 8B catalogue 14. Adapter protein: Q91ZP3.

Rh_24196: informal. “Swapping one of diaminopentanoate’s amino groups for a ketone while releasing ammonia, powered by NAD(+)”.

Best-positive ranks: adapter 2; true Horizyn 706; 8B catalogue 22. Adapter protein: E3PY99.

Rh_46184: plain. “This reaction attaches 3-methylbutanoate to CoA to make 3-methylbutanoyl-CoA, using energy from ATP.”

Best-positive ranks: adapter 2; true Horizyn 315; 8B catalogue 293. Adapter protein: Q9SEY5.

Rh_14021: plain. “Ecdysone gains a hydroxyl group, becoming 20-hydroxyecdysone, using oxygen and a reducing partner.”

Best-positive ranks: adapter 2; true Horizyn 305; 8B catalogue 311. Adapter protein: Q9VUF8.